CHAPTER 6

ENVIRONMENT AND NATURAL RESOURCES

CHAPTER ESSENCE AT A GLANCE:
Examine the political economy of the environment, governance of the Global Commons, the landmark 1992 Rio Earth Summit, the principle of Common But Differentiated Responsibilities, Kyoto Protocol commitments, resource conflicts over freshwater and fossil fuels, and the rights of indigenous communities.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Governance of Global Commons:** Areas outside sovereign national jurisdiction, including the earth's atmosphere, Antarctica, the ocean floor, and outer space, governed by multilateral regimes like the 1959 Antarctic Treaty and 1987 Montreal Protocol.

• **The 1992 Rio Earth Summit Paradigm:** Convened in Rio de Janeiro, UNCED generated Agenda 21, the Rio Declaration on Environment and Development, the Convention on Biological Diversity, and the foundational UNFCCC framework.

• **Common But Differentiated Responsibilities (CBDR):** Establishes that while all nations share responsibility for the ecological health of the planet, developed nations bear greater historical responsibility due to past industrial greenhouse emissions.

• **Geopolitics of Strategic Resources:** Critical natural resources, particularly petroleum reserves in the Middle East and transnational freshwater river basins, act as persistent flashpoints for interstate competition and strategic security doctrines.

ESSENTIAL CONSTITUTIONAL ARTICLES & THEORIES

$$\text{Global Commons Domain} \iff \{\text{Earth Atmosphere},\quad \text{Antarctica},\quad \text{Ocean Floor},\quad \text{Outer Space}\}$$

Four Extraterritorial Domains of Common Planetary Heritage

$$\text{Rio Earth Summit (1992)} \iff \text{Agenda 21} + \text{Rio Declaration} + \text{UNFCCC Climate Convention}$$

Multilateral Outcomes of the UN Conference on Environment and Development

$$\text{CBDR Principle} \iff \frac{\text{Historical Greenhouse Emissions (Developed)}}{\text{Equitable Carbon Budget and Growth Space (Developing)}}$$

Differentiated Responsibility Based on Historical Ecological Debt

$$\text{Kyoto Protocol Structure (1997)} \iff \text{Annex I Binding Emission Reductions} \rightarrow \text{Common But Differentiated Roles}$$

Binding Carbon Targets for Industrialized Nations under UNFCCC

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Do not state that developing nations were assigned mandatory emission reduction targets under the 1997 Kyoto Protocol; only industrialized Annex I nations were bound.

• **Alert:** Remember that Agenda 21 is a non-binding, voluntary action plan for sustainable development, not a punitive treaty.

• **Alert:** Avoid writing that India does not participate in climate agreements; India ratified the UNFCCC, Kyoto Protocol, and Paris Agreement, maintaining substantial renewable energy commitments.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. What is meant by the 'Global Commons'? Give two examples of domains classified under this category. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Definition:** Areas and resources of the earth that do not fall under the sovereign jurisdiction of any single state and require common international governance. **[1 Mark]**

• **Two Examples:** The earth's atmosphere and the continent of Antarctica (or the ocean floor and outer space). **[1 Mark]**

2. Mention two major institutional outcomes of the 1992 United Nations Conference on Environment and Development held in Rio de Janeiro. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Agenda 21:** A comprehensive global blueprint and voluntary action plan outlining strategies for sustainable ecological development into the 21st century. **[1 Mark]**

• **UNFCCC Framework:** The adoption of the United Nations Framework Convention on Climate Change to combat dangerous human interference with the climate system. **[1 Mark]**

3. Explain the principle of 'Common But Differentiated Responsibilities' (CBDR) in international environmental agreements. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Common Responsibility:** Recognizes that preserving the global environment is a collective duty of all states, as ecological damage affects all humanity. **[1 Mark]**

• **Differentiated Duty based on Historical Emissions:** Acknowledges that developed nations have contributed historically the largest share of greenhouse gases and possess greater financial capacity to act. **[1 Mark]**

• **Developmental Space for the Global South:** Developing countries need policy space and financial assistance to eradicate poverty before assuming stringent binding mitigation burdens. **[1 Mark]**

4. Discuss the objectives and principal provisions of the 1997 Kyoto Protocol. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Primary Objective:** An international treaty linked to the UNFCCC setting legally binding greenhouse gas reduction targets for developed countries to stabilize global warming. **[1 Mark]**

• **Annex I Obligations:** Industrialized nations committed to cut their collective emissions by an average of 5 percent below 1990 levels during the first commitment period. **[1 Mark]**

• **Exemption for Developing Nations:** Developing nations like India and China were exempted from mandatory cuts to prioritize their socio-economic development and poverty reduction. **[1 Mark]**

5. How do access to petroleum reserves and freshwater resources generate geopolitical conflicts in global politics? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Oil and Strategic Hegemony:** The concentration of petroleum reserves in the Persian Gulf has driven military interventions, foreign military bases, and diplomatic pressure by major powers to secure sea lanes. **[1 Mark]**

• **Freshwater Scarcity and Riparian Disputes:** Upstream river damming and water diversion create intense friction between upstream and downstream riparian states, as seen along the Euphrates, Tigris, and Nile rivers. **[1 Mark]**

• **Water Wars Hypothesis:** Geopolitical analysts warn that severe regional freshwater depletion may trigger interstate military skirmishes over shared transnational aquifers. **[1 Mark]**

6. Analyze the rights and environmental struggles of indigenous peoples in the context of contemporary global governance. **[Score: 4 Marks]**

Answer & Key Valuation Steps:

• **Definition of Indigenous Peoples:** Descendants of the original inhabitants of a territory who were conquered or colonized, maintaining distinct customary social and economic practices. **[1 Mark]**

• **Threat to Ancestral Habitats:** Industrial mining, massive hydroelectric dams, commercial deforestation, and agribusiness relentlessly displace indigenous tribes from their sacred territories. **[1 Mark]**

• **Traditional Ecological Knowledge:** Indigenous communities possess invaluable sustainable knowledge for preserving biodiversity and living in balance with ecosystems. **[1 Mark]**

• **Global Recognition and Demands:** The UN Declaration on the Rights of Indigenous Peoples (2007) affirmed their right to self-determination, control over ancestral lands, and free, prior, and informed consent. **[1 Mark]**

7. Outline India's perspective and commitments regarding international climate change negotiations. **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• **Low Historical and Per Capita Emissions:** India points out that its per capita greenhouse emissions are far below the global average and a fraction of developed country figures. **[1 Mark]**

• **Championing Climate Equity:** India insists that developed nations must fulfill their promised climate financing and technology transfers to assist developing countries. **[1 Mark]**

• **Proactive Domestic Initiatives:** India ratified the Paris Agreement and launched ambitious renewable programs, including the National Solar Mission and the International Solar Alliance. **[1 Mark]**